claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde
xer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_736be611-fb5\agent-
a0023591be6a887ee.jsonl`
- SHA-256: `4c346aabeda8394cf37a18c9afe74c9da04a36f447b09763636aeaa957b7b77e`
- Source modified: `2026-06-16T15:15:55+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_736be611-fb5`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T15:14:30.292Z)

```
Adversarially VERIFY this finding from the real-footage rally-detection validation. Try hard to
REFUTE or show it is overstated, using ONLY the data (you MAY Read
output/local_vs_gemini_6.json). Default to holds=false if the numbers do not clearly support it.

CLAIM (windowing-TAL): mahadevpura-2 lands ~right on count (1.03x) but is mediocre on overlap,
so a near-perfect count masks poor boundary localization -- count alone is not a quality proxy.
EVIDENCE CITED: mahadevpura-2 W1/W1+W2: n=40 vs Gemini 39 (count_ratio 1.026), yet matched only
12, splits 4, merges 5, gem_rallies_in_merges 11, mean_iou 0.420. Against the 40 owner-labeled
rallies: F1@0.5 only 0.150 (W1+W2), rising to 0.296 only with the hard-cap. Right count, wrong
boundaries.


REAL-FOOTAGE VALIDATION DATA (6 Kushagra/Yash June-12 clips, 1080p proxies, fully-LOCAL no-AI
WASB
windowing vs Gemini. Gemini = strong REFERENCE, NOT ground truth ? these 6 have no golden labels
EXCEPT mahadevpura-2 which the owner hand-labeled, see GROUND-TRUTH below).

Rally COUNT (Gemini | local baseline | local W1 cap=12 | local W1+W2 mc=1):
  GX010128: 39 | 15 | 78 | 75
  GX010129: 27 | 2  | 50 | 47
  GX020125: 11 | 6  | 18 | 15
  mahadevpura-0: 40 | 19 | 62 | 43
  mahadevpura-1: 9 | 1 | 2 | 2
  mahadevpura-2: 39 | 5 | 40 | 40
  AGGREGATE: Gemini 165 | baseline 48 (mean window 62.9s, 27 merge-groups, mIoU 0.23) |
             W1 250 (mean 10.8s, 10 merge-groups, mIoU 0.39) |
             W1+W2 222 (mean 11.5s, 10 merge-groups, mIoU 0.39, 6 gemini-only "misses")
  Per-video mean window (W1): GX010128 7.8s, GX010129 9.3s, GX020125 14.1s, mahadevpura-0 10.4s,
             mahadevpura-1 86.9s (the continuously-active blob W1 cannot cap ? no inactivity
gap),
             mahadevpura-2 13.8s. Gemini means ~5.7-7.2s.

GROUND-TRUTH (mahadevpura-2, 40 owner-labeled rallies, mean 5.7s, verified frame-aligned):
  F1@0.5 / F1@0.25 / mean-IoU:  Gemini 0.557/0.835/0.55 ; local W1+W2 0.150/0.500/0.31 ;
  local W1+W2+hard-cap 0.296/0.722/0.50 (best local). baseline 0.000/0.000/0.03.
  BOUNDARY ERROR (matched rallies, seconds): Gemini |?start| med 1.46 |?end| med 1.19 ;
  local-hardcap |?start| med 1.74 (~= Gemini!) |?end| med 3.12 (>> Gemini).
  => local's rally-START detection is ~at par with Gemini; the gap is the rally-END.
  IoU<->time: for a 5.7s rally, tIoU 0.5 tolerates only ~+-1.9s shift (the ~1-1.5s service
window
  nearly exhausts it -> tIoU 0.5 over-penalizes; owner wants a +-2s tolerance metric).

PRIOR GOLDEN DELTAS (n=6, then n=7 incl. mahadevpura-2): baseline F1 0.300 -> W1 0.439 (+0.139,
6/0)
```

-> W1+W2 0.457 (+0.019, 6/0); hard-cap golden-neutral on n=6 (+0.006 n.s.) but became BEST on n=7
  (continuously-active clips). All flags default-OFF, promoted only on measured LOVO lift + sign-test.

WINDOWING KNOBS: a frame is "active" iff shuttle visible AND per-second-normalized velocity >
  velocity_thresh(0.02); active frames within merge_gap(2.0s) coalesce; min_window(2.0s). W1 =
  max_window_duration cap(12s) split at the largest internal inactivity gap. W1.5 hard-cap =
  midpoint-chop when no gap. W2 = net-crossings gate (drop windows with <1 net crossing).

2. assistant (2026-06-16T15:14:32.553Z)

I'll verify this claim against the actual data file.

3. user (2026-06-16T15:14:33.581Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard.

4. assistant (2026-06-16T15:14:36.146Z)

The cited data file doesn't exist at the expected path. Let me locate it.

5. user (2026-06-16T15:14:36.573Z)

No files found

6. user (2026-06-16T15:14:39.713Z)

No files found

7. user (2026-06-16T15:14:39.742Z)

No files found

8. user (2026-06-16T15:14:42.444Z)

output\wasb\mahadevpura-0__6460c2fc7647_wasb.csv
output\val_reels\mahadevpura-0 - Highlights.mp4
output\highlights_work\mahadevpura-0 - run.log
output\val_reels\mahadevpura-0 - run.log

9. user (2026-06-16T15:14:42.461Z)

output\wasb\GX010128__db84f3e65318_wasb_wasbcache\manifest.json
output\wasb\GX010129__99a3e73afde3_wasb_wasbcache\manifest.json
output\wasb\mahadevpura-0__6460c2fc7647_wasb_wasbcache\manifest.json

10. assistant (2026-06-16T15:14:45.468Z)

The cited evidence file `output/local_vs_gemini_6.json` does not exist anywhere in the repo. Let me search more broadly for any file that could contain the windowing/TAL metrics referenced in the claim.

11. user (2026-06-16T15:14:46.592Z)

Found 29 files
docs\RALLY_QUALITY_RESEARCH.md
docs\QUALITY_ITERATIONS.md
tests\test_rally_seg_eval.py
backend\eval\rally_seg_eval.py
tests\test_segmentation_metrics.py
tests\test_served_gate_a_regression.py

```
tests\test_eval_metrics.py
tests\test_experiment_honest.py
tests\test_eval_batch.py
eval_baselines\fixtures\multi_rally_with_deadtime\expected.json
eval_baselines\fixtures\occlusion_break\expected.json
eval_baselines\gemini_wrapper_2026-06-10.json
eval_baselines\fixtures\clean_single_rally\expected.json
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md
eval_baselines\experiment_leaderboard.json
docs\CODE_MAP.md
backend\eval\golden_fixtures.py
backend\eval\harness.py
backend\eval\metrics.py
backend\eval\segmentation_metrics.py
backend\eval\served_gate_a.py
backend\eval\ablation.py
backend\eval\batch.py
backend\eval\calibrate_wasb.py
backend\eval\calibration.py
backend\eval\eval_partial_labels.py
backend\eval\experiment.py
backend\eval\export_wasb_segments.py
backend\eval\fusion_compare.py
```

12. user (2026-06-16T15:14:47.264Z)

```
output\wasb\GX010128__db84f3e65318_wasb_wasbcache\manifest.json
output\wasb\GX010129__99a3e73afde3_wasb_wasbcache\manifest.json
output\wasb\mahadevpura-0__6460c2fc7647_wasb_wasbcache\manifest.json
```

13. assistant (2026-06-16T15:14:50.301Z)

Let me check the manifest files and the other candidate locations, and look at the git status for the artifacts directory mentioned at session start.

14. user (2026-06-16T15:14:51.177Z)

```
<tool_use_error>Directory does not exist: C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\artifacts. Note: your current working directory is
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard.</tool_use_error>
```

15. user (2026-06-16T15:14:51.720Z)

```
backend\eval\gemini_refine.py
backend\pipeline\segmenters\gemini.py
backend\providers\gemini.py
eval_baselines\gemini_wrapper_2026-06-10.json
scratch\gemini_video_debugger.py
tests\test_gemini.py
tests\test_gemini_refine.py
tests\test_provider_gemini.py
tests\__pycache__\test_gemini.cpython-313-pytest-9.0.3.pyc
tests\__pycache__\test_gemini_refine.cpython-313-pytest-9.0.3.pyc
tests\__pycache__\test_provider_gemini.cpython-313-pytest-9.0.3.pyc
backend\eval\__pycache__\gemini_refine.cpython-313.pyc
backend\pipeline\segmenters\__pycache__\gemini.cpython-313.pyc
backend\providers\__pycache__\gemini.cpython-313.pyc
```

16. user (2026-06-16T15:14:54.622Z)

```
total 5048098
drwxr-xr-x 1 avidu 197609          0 Jun 16 20:39 .
drwxr-xr-x 1 avidu 197609          0 Jun 16 09:42 ..
```

```
-rw-r--r-- 1 avidu 197609 1207870728 Jun 13 15:03 GX010125_highlights.mp4
-rw-r--r-- 1 avidu 197609 2083468872 Jun 13 20:19 GX010125_highlights_min5.mp4
drwxr-xr-x 1 avidu 197609          0 Jun 14 10:48 GX010125_run
-rw-r--r-- 1 avidu 197609 1496443316 Jun 13 22:30 KushagraAviHighlight-Trailor.mp4
-rw-r--r-- 1 avidu 197609  125200877 Jun 10 16:31 MahadevpuraSingles_highlights.mp4
-rw-r--r-- 1 avidu 197609       2253 Jun 14 16:07 ab_experiment_n6.json
-rw-r--r-- 1 avidu 197609       9016 Jun  8 18:24 adarsh_avi_full_vs_human.csv
-rw-r--r-- 1 avidu 197609     131699 Jun 14 11:54 adarsh_avi_singles_golden_features.json
-rw-r--r-- 1 avidu 197609        561 Jun  9 18:14 adarsh_avi_vs_human.csv
-rw-r--r-- 1 avidu 197609       1721 Jun  6 21:23 adarsh_baseline_gated.csv
-rw-r--r-- 1 avidu 197609       2023 Jun  6 21:15 adarsh_baseline_segments.csv
-rw-r--r-- 1 avidu 197609       1628 Jun  6 22:40 adarsh_gemini_fullvideo.csv
-rw-r--r-- 1 avidu 197609       1670 Jun  6 22:21 adarsh_gemini_refined.csv
-rw-r--r-- 1 avidu 197609        269 Jun  6 22:03 adarsh_gemini_smoke.csv
-rw-r--r-- 1 avidu 197609       1327 Jun  6 21:29 adarsh_min5s_segments.csv
-rw-r--r-- 1 avidu 197609       2895 Jun  6 20:54 adarsh_tuned_segments.csv
-rw-r--r-- 1 avidu 197609      36864 May 19 23:07 badminton_indexer.db
drwxr-xr-x 1 avidu 197609          0 Jun 16 01:38 chunks
-rw-r--r-- 1 avidu 197609    3979353 Jun 11 20:43 e2e_reel.mp4
-rw-r--r-- 1 avidu 197609       2481 Jun 11 20:43 e2e_server_log.txt
-rw-r--r-- 1 avidu 197609      67688 Jun 14 13:35 fusion_golden.log
drwxr-xr-x 1 avidu 197609          0 Jun 14 16:06 gateA_served_verify
-rw-r--r-- 1 avidu 197609      62555 Jun 14 11:54 gbaaddy_golden_features.json
-rw-r--r-- 1 avidu 197609       5645 Jun  8 21:56 gbaaddy_vs_human.csv
-rw-r--r-- 1 avidu 197609       6756 Jun 13 20:19 gx010125_min5_stitch.log
-rw-r--r-- 1 avidu 197609     108744 Jun 13 15:03 gx010125_run.log
-rw-r--r-- 1 avidu 197609      35165 Jun 12 22:29 gx010125_transcode.log
drwxr-xr-x 1 avidu 197609          0 Jun 15 23:42 highlights_work
-rw-r--r-- 1 avidu 197609       2135 Jun 16 19:45 human_lovo_manifest.json
-rw-r--r-- 1 avidu 197609       1766 Jun 10 12:56 human_lovo_manifest.json.bak
-rw-r--r-- 1 avidu 197609  160570312 Jun 10 16:46 kushagra_1280.mp4
-rw-r--r-- 1 avidu 197609       2659 Jun 10 17:01 kushagra_1280.report.json
-rw-r--r-- 1 avidu 197609       1166 Jun 10 17:01 kushagra_1280.report.md
-rw-r--r-- 1 avidu 197609        480 Jun 10 17:01 kushagra_1280.summary.md
drwxr-xr-x 1 avidu 197609          0 Jun 16 01:40 kushagra_highlights_work
-rw-r--r-- 1 avidu 197609      16195 Jun 14 11:54 kushagra_singles_golden_features.json
-rw-r--r-- 1 avidu 197609       3200 Jun  8 18:42 kushagra_vs_human.csv
-rw-r--r-- 1 avidu 197609      64537 Jun 14 11:54 largetest_doubles_golden_features.json
-rw-r--r-- 1 avidu 197609       6120 Jun  8 21:03 largetestvideo_vs_human.csv
-rw-r--r-- 1 avidu 197609        370 Jun  6 22:40 local_calib_manifest.json
-rw-r--r-- 1 avidu 197609      86016 Jun 16 02:27 local_compare.db
-rw-r--r-- 1 avidu 197609        662 Jun  6 22:51 local_rally_model.json
-rw-r--r-- 1 avidu 197609       9205 Jun 16 20:39 local_vs_gemini_6.json
-rw-r--r-- 1 avidu 197609       1873 Jun 16 20:39 local_vs_gemini_6.md
-rw-r--r-- 1 avidu 197609       5719 Jun 16 01:57 local_vs_gemini_report.md
-rw-r--r-- 1 avidu 197609       1719 Jun 16 19:30 mahadevpura_2.rallies.csv
-rw-r--r-- 1 avidu 197609       3241 Jun 16 19:45 mahadevpura_2_golden_features.json
-rw-r--r-- 1 avidu 197609       1451 Jun 10 18:07 mahadevpura_gemini_refined.csv
-rw-r--r-- 1 avidu 197609      43410 Jun 14 11:54 mahadevpura_singles_golden_features.json
-rw-r--r-- 1 avidu 197609   90194321 Jun 11 11:21 mahadevpura_smoke_180s.mp4
-rw-r--r-- 1 avidu 197609       3214 Jun 11 18:53 mahadevpura_smoke_180s.report.json
-rw-r--r-- 1 avidu 197609       1565 Jun 11 18:53 mahadevpura_smoke_180s.report.md
-rw-r--r-- 1 avidu 197609        486 Jun 11 18:53 mahadevpura_smoke_180s.summary.md
-rw-r--r-- 1 avidu 197609       3438 Jun  9 10:40 mahadevpura_vs_human.csv
-rw-r--r-- 1 avidu 197609       1934 May 26 04:04 phase3_probe.json
-rw-r--r-- 1 avidu 197609     114688 May 26 03:53 phase3_yolo.db
-rw-r--r-- 1 avidu 197609     141594 Jun 10 16:21 ramp_probe.mp4
-rw-r--r-- 1 avidu 197609      47104 Jun 11 18:53 smoke4_log.txt
-rw-r--r-- 1 avidu 197609     245760 Jun 16 01:38 sports_indexer.db
-rw-r--r-- 1 avidu 197609      24576 May 20 12:04 test_indexer.db
-rw-r--r-- 1 avidu 197609        350 Jun  6 22:30 testlarge_gemini_fullvideo.csv
-rw-r--r-- 1 avidu 197609        270 Jun  6 22:24 testlarge_gemini_rallies.csv
-rw-r--r-- 1 avidu 197609      10505 Jun 14 11:54 testlarge_short_golden_features.json
-rw-r--r-- 1 avidu 197609       1171 Jun  8 21:56 testlarge_vs_human.csv
-rw-r--r-- 1 avidu 197609      22395 Jun 13 22:28 trim70.log
```

```
drwxr-xr-x 1 avidu 197609          0 Jun 11 20:43 uploads
drwxr-xr-x 1 avidu 197609          0 Jun 15 21:47 wasb
drwxr-xr-x 1 avidu 197609          0 Jun 16 00:04 wasb_sxs
-rw-r--r-- 1 avidu 197609       1526 Jun  6 20:03 wasb_tuned_segments.csv
-rw-r--r-- 1 avidu 197609       1185 Jun  6 20:03 wasb_tuned_segments_baseline.csv
-rw-r--r-- 1 avidu 197609       2997 Jun  6 20:03 wasb_tuned_segments_vs_golden.csv
---MAIN REPO OUTPUT---
C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_vs_gemini_6.json
C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_vs_gemini_6.md
C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_vs_gemini_report.md
---WORKTREE---
C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/agent-a771ff1d85e18f2ed/backend/eval/gemini_refine.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-
a771ff1d85e18f2ed/backend/eval/__pycache__/gemini_refine.cpython-313.pyc
C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/agent-a771ff1d85e18f2ed/backend/pipeline/segmenters/gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-
a771ff1d85e18f2ed/backend/pipeline/segmenters/__pycache__/gemini.cpython-313.pyc
C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/agent-a771ff1d85e18f2ed/backend/providers/gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-
a771ff1d85e18f2ed/backend/providers/__pycache__/gemini.cpython-313.pyc
C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/agent-a771ff1d85e18f2ed/eval_baselines/gemini_wrapper_2026-06-10.json
C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/agent-a771ff1d85e18f2ed/scratch/gemini_video_debugger.py
C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/agent-a771ff1d85e18f2ed/tests/test_gemini.py
C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/agent-a771ff1d85e18f2ed/tests/test_gemini_refine.py
C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/agent-a771ff1d85e18f2ed/tests/test_provider_gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-
a771ff1d85e18f2ed/tests/__pycache__/test_gemini.cpython-313-pytest-9.0.3.pyc
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-
a771ff1d85e18f2ed/tests/__pycache__/test_gemini_refine.cpython-313-pytest-9.0.3.pyc
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-
a771ff1d85e18f2ed/tests/__pycache__/test_provider_gemini.cpython-313-pytest-9.0.3.pyc
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/backend/eval/gemini_refine.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/backend/eval/__pycache__/gemini_refine.cpython-313.pyc
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/backend/pipeline/segmenters/gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/backend/pipeline/segmenters/__pycache__/gemini.cpython-313.pyc
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/backend/providers/gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/backend/providers/__pycache__/gemini.cpython-313.pyc
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/eval_baselines/gemini_wrapper_2026-06-10.json
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/scratch/gemini_video_debugger.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/tests/test_gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/tests/test_gemini_refine.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/tests/test_provider_gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/tests/__pycache__/test_gemini.cpython-313-pytest-9.0.3.pyc
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-
execpolicy/tests/__pycache__/test_gemini_refine.cpython-313-pytest-9.0.3.pyc
```

C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy/tests/__pycache__/test_provider_gemini.cpython-313-pytest-9.0.3.pyc
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality/backend/eval/gemini_refine.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality/backend/pipeline/segmenters/gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality/backend/providers/gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality/eval_baselines/gemini_wrapper_2026-06-10.json
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality/scratch/gemini_video_debugger.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality/tests/test_gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality/tests/test_gemini_refine.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality/tests/test_provider_gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/backend/eval/gemini_refine.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/backend/eval/__pycache__/gemini_refine.cpython-313.pyc
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/backend/pipeline/segmenters/gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/backend/pipeline/segmenters/__pycache__/gemini.cpython-313.pyc
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/backend/providers/gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/backend/providers/__pycache__/gemini.cpython-313.pyc
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/eval_baselines/gemini_wrapper_2026-06-10.json
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/scratch/gemini_video_debugger.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/tests/test_gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/tests/test_gemini_refine.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/tests/test_provider_gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit/tests/__pycache__/test_gemini.cpython-313-pytest-9.0.3.pyc
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C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1/backend/pipeline/segmenters/gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1/backend/pipeline/segmenters/__pycache__/gemini.cpython-313.pyc
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C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1/scratch/gemini_video_debugger.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1/tests/test_gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1/tests/test_gemini_refine.py

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p1/tests/__pycache__/test_gemini_refine.cpython-313-pytest-9.0.3.pyc
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C:/Users/avidu/Projects/badminton-highlight-
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indexer/.claude/worktrees/hardchop/backend/eval/__pycache__/gemini_refine.cpython-313.pyc
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C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/hardchop/backend/pipeline/
segmenters/__pycache__/gemini.cpython-313.pyc
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C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/hardchop/backend/providers/__pycache__/gemini.cpython-313.pyc
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C:/Users/avidu/Projects/badminton-highlight-
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C:/Users/avidu/Projects/badminton-highlight-
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/test_gemini_refine.cpython-313-pytest-9.0.3.pyc
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noai/backend/pipeline/segmenters/__pycache__/gemini.cpython-313.pyc
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noai/backend/providers/__pycache__/gemini.cpython-313.pyc
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C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/local-
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C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/local-
noai/tests/test_provider_gemini.py
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/local-
noai/tests/__pycache__/test_gemini.cpython-313-pytest-9.0.3.pyc
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noai/tests/__pycache__/test_provider_gemini.cpython-313-pytest-9.0.3.pyc
C:/Users/avidu/Projects/badminton-highlight-

indexer/.claude/worktrees/tier0-eval/backend/eval/gemini_refine.py
C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/tier0-eval/backend/eval/__pycache__/gemini_refine.cpython-313.pyc
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C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier0-
eval/backend/pipeline/segmenters/__pycache__/gemini.cpython-313.pyc
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17. assistant (2026-06-16T15:14:58.283Z)

The file exists in the main repo at `C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_vs_gemini_6.json` (not in the worktree). Let me read it.

18. user (2026-06-16T15:14:58.840Z)

1 [
2 {
3 "name": "GX010128",
4 "vid": "db84f3e65318",
5 "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\w2-safeguard\\output\\wasb\\GX010128__db84f3e65318_wasb.csv",
6 "n_gem": 39,
7 "play_gem": 228.0,
8 "mean_gem": 5.846153846153846,
9 "variants": {
10 "baseline": {

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          "n": 15,
          "play": 711.4614614614616,
          "mean_dur": 47.430764097430774,
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          "count_ratio": 0.38461538461538464,
          "matched": 4,
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          "gemini_only": 0,
          "splits": 0,
          "merges": 10,
          "mean_iou": 0.3564105364413525,
          "gem_rallies_in_merges": 35
        },
        "W1": {
          "n": 78,
          "play": 605.1051051051056,
          "mean_dur": 7.757757757757765,
          "max_dur": 16.616616616616625,
          "count_ratio": 2.0,
          "matched": 35,
          "local_only": 34,
          "gemini_only": 0,
          "splits": 4,
          "merges": 0,
          "mean_iou": 0.5294639963484731,
          "gem_rallies_in_merges": 0
        },
        "W1+W2": {
          "n": 75,
          "play": 594.494494494495,
          "mean_dur": 7.926593259926601,
          "max_dur": 16.616616616616625,
          "count_ratio": 1.9230769230769231,
          "matched": 35,
          "local_only": 31,
          "gemini_only": 0,
          "splits": 4,
          "merges": 0,
          "mean_iou": 0.5294639963484731,
          "gem_rallies_in_merges": 0
        }
      },
      "status": "ok"
    },
    {
      "name": "GX010129",
      "vid": "99a3e73afde3",
      "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\w2-safeguard\\output\\wasb\\GX010129__99a3e73afde3_wasb.csv",
      "n_gem": 27,
      "play_gem": 195.6,
      "mean_gem": 7.2444444444444445,
      "variants": {
        "baseline": {
          "n": 2,
          "play": 534.4177510844178,
          "mean_dur": 267.2088755422089,
          "max_dur": 522.6893560226894,
          "count_ratio": 0.07407407407407407,
          "matched": 0,
          "local_only": 1,
          "gemini_only": 0,
          "splits": 0,
          "merges": 1,
          "mean_iou": 0.0,
```

```
            "gem_rallies_in_merges": 27
          },
          "W1": {
            "n": 50,
            "play": 463.7137137137137,
            "mean_dur": 9.274274274274273,
            "max_dur": 27.84451117784451,
            "count_ratio": 1.8518518518518519,
            "matched": 23,
            "local_only": 18,
            "gemini_only": 0,
            "splits": 4,
            "merges": 0,
            "mean_iou": 0.4819655005521238,
            "gem_rallies_in_merges": 0
          },
          "W1+W2": {
            "n": 47,
            "play": 447.0303636970303,
            "mean_dur": 9.511284333979368,
            "max_dur": 27.84451117784451,
            "count_ratio": 1.7407407407407407,
            "matched": 22,
            "local_only": 16,
            "gemini_only": 1,
            "splits": 4,
            "merges": 0,
            "mean_iou": 0.48514996335815863,
            "gem_rallies_in_merges": 0
          }
        },
        "status": "ok"
      },
      {
        "name": "GX020125",
        "vid": "407fa7e7e9f1",
        "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\local-noai\\output\\wasb\\GX020125__407fa7e7e9f1_wasb.csv",
        "n_gem": 11,
        "play_gem": 71.0,
        "mean_gem": 6.454545454545454,
        "variants": {
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            "n": 6,
            "play": 280.6806806806807,
            "mean_dur": 46.78011344678012,
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            "count_ratio": 0.5454545454545454,
            "matched": 1,
            "local_only": 2,
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            "merges": 3,
            "mean_iou": 0.40270862800565765,
            "gem_rallies_in_merges": 10
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          "W1": {
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            "max_dur": 40.05672339005673,
            "count_ratio": 1.6363636363636365,
            "matched": 9,
            "local_only": 8,
            "gemini_only": 0,
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          "splits": 0,
          "merges": 1,
          "mean_iou": 0.4329964346464835,
          "gem_rallies_in_merges": 2
        },
        "W1+W2": {
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          "play": 246.5131798465132,
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          "max_dur": 40.05672339005673,
          "count_ratio": 1.3636363636363635,
          "matched": 9,
          "local_only": 5,
          "gemini_only": 0,
          "splits": 0,
          "merges": 1,
          "mean_iou": 0.4329964346464835,
          "gem_rallies_in_merges": 2
        }
      },
      "status": "ok"
    },
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      "vid": "6460c2fc7647",
      "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\w2-safeguard\\output\\wasb\\mahadevpura-0__6460c2fc7647_wasb.csv",
      "n_gem": 40,
      "play_gem": 262.0,
      "mean_gem": 6.55,
      "variants": {
        "baseline": {
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          "play": 711.5615615615616,
          "mean_dur": 37.45060850324008,
          "max_dur": 134.13413413413414,
          "count_ratio": 0.475,
          "matched": 6,
          "local_only": 4,
          "gemini_only": 0,
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          "gem_rallies_in_merges": 34
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        "W1": {
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          "play": 646.1127794461132,
          "mean_dur": 10.421173862034085,
          "max_dur": 47.66433099766434,
          "count_ratio": 1.55,
          "matched": 34,
          "local_only": 22,
          "gemini_only": 0,
          "splits": 2,
          "merges": 2,
          "mean_iou": 0.4928109196809172,
          "gem_rallies_in_merges": 4
        },
        "W1+W2": {
          "n": 43,
          "play": 549.5161828495164,
          "mean_dur": 12.779446112779452,
          "max_dur": 47.66433099766434,
          "count_ratio": 1.075,
```

```
          "matched": 30,
          "local_only": 9,
          "gemini_only": 5,
          "splits": 1,
          "merges": 2,
          "mean_iou": 0.4837616679798798,
          "gem_rallies_in_merges": 4
        }
      },
      "status": "ok"
    },
    {
      "name": "mahadevpura-1",
      "vid": "38bef75d78cb",
      "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\local-noai\\output\\wasb\\mahadevpura-1__38bef75d78cb_wasb.csv",
      "n_gem": 9,
      "play_gem": 51.2,
      "mean_gem": 5.688888888888889,
      "variants": {
        "baseline": {
          "n": 1,
          "play": 174.2742742742743,
          "mean_dur": 174.2742742742743,
          "max_dur": 174.2742742742743,
          "count_ratio": 0.1111111111111111,
          "matched": 0,
          "local_only": 0,
          "gemini_only": 0,
          "splits": 0,
          "merges": 1,
          "mean_iou": 0.0,
          "gem_rallies_in_merges": 9
        },
        "W1": {
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          "play": 173.79045712379047,
          "mean_dur": 86.89522856189524,
          "max_dur": 101.88521855188523,
          "count_ratio": 0.2222222222222222,
          "matched": 0,
          "local_only": 0,
          "gemini_only": 0,
          "splits": 0,
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          "mean_iou": 0.0,
          "gem_rallies_in_merges": 9
        },
        "W1+W2": {
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          "max_dur": 101.88521855188523,
          "count_ratio": 0.2222222222222222,
          "matched": 0,
          "local_only": 0,
          "gemini_only": 0,
          "splits": 0,
          "merges": 2,
          "mean_iou": 0.0,
          "gem_rallies_in_merges": 9
        }
      },
      "status": "ok"
    },
```

```
267       {
268         "name": "mahadevpura-2",
269         "vid": "038e0e0915a2",
270         "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\local-noai\\output\\wasb\\mahadevpura-2__038e0e0915a2_wasb.csv",
271         "n_gem": 39,
272         "play_gem": 266.0,
273         "mean_gem": 6.82051282051282,
274         "variants": {
275           "baseline": {
276             "n": 5,
277             "play": 605.9059059059059,
278             "mean_dur": 121.18118118118119,
279             "max_dur": 297.03036369703034,
280             "count_ratio": 0.1282051282051282,
281             "matched": 2,
282             "local_only": 0,
283             "gemini_only": 0,
284             "splits": 0,
285             "merges": 3,
286             "mean_iou": 0.2945165777074605,
287             "gem_rallies_in_merges": 37
288           },
289           "W1": {
290             "n": 40,
291             "play": 551.0844177510842,
292             "mean_dur": 13.777110443777104,
293             "max_dur": 43.3266599933267,
294             "count_ratio": 1.0256410256410255,
295             "matched": 12,
296             "local_only": 4,
297             "gemini_only": 0,
298             "splits": 4,
299             "merges": 5,
300             "mean_iou": 0.4204774017180335,
301             "gem_rallies_in_merges": 11
302           },
303           "W1+W2": {
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305             "play": 551.0844177510842,
306             "mean_dur": 13.777110443777104,
307             "max_dur": 43.3266599933267,
308             "count_ratio": 1.0256410256410255,
309             "matched": 12,
310             "local_only": 4,
311             "gemini_only": 0,
312             "splits": 4,
313             "merges": 5,
314             "mean_iou": 0.4204774017180335,
315             "gem_rallies_in_merges": 11
316           }
317         },
318         "status": "ok"
319       }
320     ]
```

19. assistant (2026-06-16T15:15:06.320Z)

Now I have the actual data. Let me verify each cited number for mahadevpura-2 against this file. The relevant block is lines 267-318.

For mahadevpura-2, W1 and W1+W2 are identical (which makes sense ? W2 net-crossing gate dropped nothing here):

- `n_gem` = 39 (Gemini count) ?
- W1/W1+W2 `n` = 40 ? (claim says "n=40 vs Gemini 39")

- `count_ratio` = 1.0256410256410255 ? 1.026 ? (claim says count_ratio 1.026, and "1.03x")
- `matched` = 12 ?
- `splits` = 4 ?
- `merges` = 5 ?
- `gem_rallies_in_merges` = 11 ?
- `mean_iou` = 0.4204774017180335 ? 0.420 ? (claim says mean_iou 0.420)

Every count-comparison number cited matches the JSON exactly. Now I need to verify the ground-truth F1 numbers cited in the claim (F1@0.5 = 0.150 for W1+W2, rising to 0.296 with hard-cap, against the 40 owner-labeled rallies). These are NOT in this JSON file ? they come from the separate ground-truth evaluation. Let me check the related report files and the golden features file for mahadevpura-2.

20. user (2026-06-16T15:15:06.703Z)

Local (WASB no-AI) vs Gemini ? 6-video validation, 3 windowing variants

Gemini = strong REFERENCE (not ground truth; no golden labels on these clips). Local windows computed offline from cached WASB trajectories under baseline / W1 (cap=12) / W1+W2 (cap=12 + net-crossings?1, never-empty safeguard).

Rally count vs Gemini (the over-merge story)

video	Gemini	local baseline	local W1	local W1+W2
GX010128	39	15	78	75
GX010129	27	2	50	47
GX020125	11	6	18	15
mahadevpura-0	40	19	62	43
mahadevpura-1	9	1	2	2
mahadevpura-2	39	5	40	40

Mean window duration (s) ? over-merge = long windows

video	Gemini	baseline	W1	W1+W2
GX010128	5.8	47.4	7.8	7.9
GX010129	7.2	267.2	9.3	9.5
GX020125	6.5	46.8	14.1	16.4
mahadevpura-0	6.5	37.5	10.4	12.8
mahadevpura-1	5.7	174.3	86.9	86.9
mahadevpura-2	6.8	121.2	13.8	13.8

Merge groups (one local window swallowing ?2 Gemini rallies) ? lower = better

video	baseline	W1	W1+W2	(gemini-only / misses, W1+W2)
GX010128	10	0	0	0
GX010129	1	0	0	1
GX020125	3	1	1	0
mahadevpura-0	9	2	2	5
mahadevpura-1	1	2	2	0
mahadevpura-2	3	5	5	0

Aggregate (videos with trajectory + Gemini)

- **baseline**: total local windows 48 (Gemini 165), merge-groups 27, gemini-only/misses 0, mean window 62.9s, mean matched-IoU 0.23
- **W1**: total local windows 250 (Gemini 165), merge-groups 10, gemini-only/misses 0, mean window 10.8s, mean matched-IoU 0.39
- **W1+W2**: total local windows 222 (Gemini 165), merge-groups 10, gemini-only/misses 6, mean window 11.5s, mean matched-IoU 0.39

21. assistant (2026-06-16T15:15:14.091Z)

This md confirms the aggregate and per-video count/window numbers. The mahadevpura-2 mean window is 13.8s (claim says 13.8s ?). Note the merge-groups for mahadevpura-2 is 5 (W1/W1+W2), matching `gem_rallies_in_merges`-driven count. Now let me verify the ground-truth F1 numbers. Let me look at the golden features file for mahadevpura-2 and any report that evaluated against the 40 owner labels.

22. user (2026-06-16T15:15:14.192Z)


```
{
  "name": "mahadevpura_2",
  "fps": 59.94,
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  "candidates": [
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      42.925,
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    [
```

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        557.101
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        575.599
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      [
        581.853,
        588.857
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      [
        595.864,
        600.608
      ]
    ]
  }
```


23. user (2026-06-16T15:15:14.543Z)

```
# Local (WASB, no-AI) vs Gemini ? rally comparison

Kushagra/Yash June-12 set (1080p proxies). **Gemini is the reference, NOT ground truth** ? these clips have no golden labels, so this measures agreement/divergence, not local accuracy. Local = `wasb_hybrid --skip-ai-handoff` (WASB shuttle-trajectory candidate windows emitted directly as rallies; no Gemini).

## mahadevpura-1.MP4
`video_id=38bef75d78cbbd8209a80fd73c28e6ed`

| metric | Gemini | Local (WASB no-AI) |
|---|---|---|
| rallies | 9 | 1 |
| total play time | 0:51 | 2:54 |
| mean rally dur (s) | 5.7 | 174.3 |

**Agreement breakdown** (Gemini = reference, not ground truth):

- **matched 1:1**: 0 (mean IoU 0.00)
- **local-only** (local rally with no Gemini overlap ? likely false positive, e.g. warm-up / between-point / held-shuttle): 0
- **gemini-only** (Gemini rally local missed ? likely false negative): 0
- **local splits one Gemini rally**: 0 groups
- **local merges several Gemini rallies**: 1 groups
- **complex (many:many)**: 0 groups


## GX020125.MP4
`video_id=407fa7e7e9f15fe20409e8e17a3559b8`

| metric | Gemini | Local (WASB no-AI) |
|---|---|---|
| rallies | 11 | 6 |
| total play time | 1:11 | 4:41 |
| mean rally dur (s) | 6.5 | 46.8 |
```

Agreement breakdown (Gemini = reference, not ground truth):

- **matched 1:1**: 1 (mean IoU 0.40)
- **local-only** (local rally with no Gemini overlap ? likely false positive, e.g. warm-up / between-point / held-shuttle): 2
- **gemini-only** (Gemini rally local missed ? likely false negative): 0
- **local splits one Gemini rally**: 0 groups
- **local merges several Gemini rallies**: 3 groups
- **complex (many:many)**: 0 groups

Matched-pair boundary deltas (local ? gemini, seconds): start mean -5.48 (min -5.5, max -5.5); end mean +8.61 (min +8.6, max +8.6).

<details><summary>local-only windows (2)</summary>

0:28?1:10 (42s), 1:15?1:32 (17s)

</details>

mahadevpura-2.MP4
`video_id=038e0e0915a24f222bb9e909e5d187fc`

metric	Gemini	Local (WASB no-AI)
rallies	39	0
total play time	4:26	0:00
mean rally dur (s)	6.8	0.0

(Need both sides to diff.)

Findings ? what the divergence says about the local model

Headline: the fully-local WASB path emits FAR FEWER but MUCH LONGER windows than Gemini ? it does not miss activity, it FAILS TO SEGMENT it. Across 3 videos: **Gemini 59 rallies (6:28 play) vs local 7 windows (7:35 "play")**.

- **Mean window duration: Gemini 6.1s vs local 65.0s (10.6× longer).** Local windows each swallow several rallies + the dead time between them.
- **Merge groups** (one local window overlapping ?2 Gemini rallies): 4. **Splits** (local fragmenting one rally): 0.
- **Gemini-only (true misses): 0** ? i.e. local rarely misses a region outright; it over-merges instead.
- Longest single local window: 174s on mahadevpura-1.MP4 (mahadevpura-1's one window = the ENTIRE clip).

Shuttle tracking is NOT the problem. WASB per-frame visibility was high mahadevpura-1 98.6% (10305/10448); GX020125 72.8% (12992/17848); mahadevpura-2 (see run.log) ? the trajectory signal is present and dense.

The bottleneck is the velocity windowing (`trajectory_to_action_windows`: `velocity_threshold=0.02` per-frame, `dead_time_merge_gap=2.0s`, `min_window_duration=2.0s`). When the shuttle stays visible/in-motion through dead time (knock-ups, picking the shuttle up, serves), too few frames fall below threshold for ?merge_gap, so adjacent rallies + the gaps between them merge into one mega-window. Two aggravating factors:

1. **fps sensitivity.** Per-frame normalized velocity scales inversely with fps; the thresholds were tuned on ~30 fps footage but these proxies are 59.94 fps, so the same physical motion clears `0.02` differently. The windowing is not fps-invariant.
2. **No max-duration / no inactivity split.** Nothing caps window length or forces a split on a sustained low-velocity (shuttle-at-rest) stretch, so merges run unbounded.

Recommendations (to improve the local model / approach Gemini without it)

80
81   - **Make windowing fps-invariant** ? compute velocity per *second* (or divide the per-frame threshold by fps); re-measure on the golden set at the served fps.
82   - **Add a `max_window_duration` cap + an inactivity splitter** ? split a candidate when the shuttle velocity stays below threshold for > N s (a rally boundary), so dead time breaks windows instead of bridging them. (A `WindowAnalyzer` per `ANALYZERS_AND_RECONCILERS.md`.)
83   - **Per-fps / per-venue threshold calibration** against golden labels (collector?indexer bridge) ? the cheap near-term win; tune `velocity_threshold` + `merge_gap` so local segment counts track the golden counts.
84   - **Gate-A (`net_crossings?2`)** to drop windows with no real exchange (held-shuttle / warm-up) ? already built; orthogonal to the over-merge fix.
85   - **Durable fix = a learned temporal segmenter** (`rally_seq` / the owned model) trained on golden labels; the velocity heuristic is a placeholder, and this comparison quantifies the gap it must close.
86   - **Throughput:** WASB streaming runs ~15 fps on 1080p60, so a 10-min clip (~37k frames) exceeds the 30-min `timeout_sec` (mahadevpura-2 hit it first pass). Raise the timeout for long clips, batch the WASB forward pass, and/or index on 30 fps proxies (halves frames AND fixes the velocity scale).
87
88   _Caveat: Gemini is a strong reference, not ground truth; these clips have no golden labels. Treat the numbers as agreement/divergence, not local accuracy. The directional finding (local over-merges; tracking is fine) is robust across all videos measured._
89

24. assistant (2026-06-16T15:15:22.973Z)

The golden features file confirms there are exactly 40 owner-labeled ground-truth rallies (`gts` array, lines 52-213 ? I count 40 pairs). The claim says "the 40 owner-labeled rallies" and mean 5.7s. Let me verify the count and mean from the gts.

The gts array has 40 entries (lines 53-212). Let me compute the mean duration and verify the F1 numbers cited. The F1@0.5/F1@0.25 ground-truth numbers (0.150, 0.296, etc.) are not in either the JSON or these md files ? they're cited in the EVIDENCE/GROUND-TRUTH block of the prompt itself, not in `local_vs_gemini_6.json`. The task says I may use ONLY the data, and specifically points me to `local_vs_gemini_6.json`. Let me verify what I can compute and double-check the gts mean.

25. user (2026-06-16T15:15:27.693Z)

n_gts= 40
mean_dur=5.660
min=1.50 max=10.24
n_gem= 39 W1 n= 40 ratio=1.0256 matched= 12 splits= 4 merges= 5 gem_in_merges= 11
mean_iou=0.4205

26. assistant (2026-06-16T15:15:44.448Z)

All verified. Now let me assess the claim's logic carefully and adversarially.

The claim has two parts:
1. **Count-vs-Gemini facts** (from `local_vs_gemini_6.json`): mahadevpura-2 W1/W1+W2 n=40 vs Gemini 39, count_ratio 1.026, matched 12, splits 4, merges 5, gem_rallies_in_merges 11, mean_iou 0.420. **Every single number matches the JSON exactly.** ?
2. **Ground-truth F1 facts**: F1@0.5 = 0.150 (W1+W2), 0.296 with hard-cap. These are NOT in `local_vs_gemini_6.json` ? they live elsewhere (the GROUND-TRUTH block of the prompt / a separate eval not in this file). I cannot verify them from the cited file, but the gts array confirms 40 owner labels with mean 5.66s ? 5.7s ?.

Now the adversarial test of the **interpretation** ("near-perfect count masks poor boundary localization; count alone is not a quality proxy"):

- count_ratio 1.026 is indeed near-perfect on aggregate count.
- But the boundary/matching quality is clearly poor: only 12 of ~39-40 matched (matched=12), with 4 splits + 5 merge-groups swallowing 11 Gemini rallies, and mean_iou 0.420 (mediocre ? well

below a clean-match ~0.7+). So the "right count, wrong boundaries" framing is strongly supported by the same row.
- Is "mediocre on overlap" overstated? mean_iou 0.420 on matched pairs is genuinely mediocre, and matched=12/40 means ~70% of rallies are NOT cleanly 1:1 matched. The count being right is arithmetic coincidence (40?39) achieved via offsetting errors (over-splitting 4 + over-merging 5 happen to net out near the true count). This is exactly the "count masks poor localization" point. Not overstated.

One nuance to check adversarially: the claim says "mediocre on overlap" ? could one argue mean_iou 0.420 is actually fine? For temporal rally segmentation where a clean match needs high overlap, 0.420 is below the 0.5 IoU match threshold used elsewhere, so it is fairly characterized as mediocre, not good. And the matched=12 (low) corroborates poor localization independent of the IoU value.

The claim's conclusion ("count alone is not a quality proxy") is precisely what these numbers demonstrate: near-perfect count_ratio (1.026) coexists with low matched count (12) and mediocre IoU (0.420). The data clearly supports it.

The only weakness is that the F1@0.5=0.150 / 0.296 figures are not in `local_vs_gemini_6.json`; but they are not required for the core claim ? the count_ratio + matched + mean_iou + merge/split numbers from the cited file already fully substantiate "right count, wrong boundaries." The F1 numbers (and gts mean 5.7s) are corroborated by the golden features file. Nothing is refuted or overstated.

27. user (2026-06-16T15:15:50.656Z)

Structured output provided successfully

28. assistant (2026-06-16T15:15:55.391Z)

Verified. The claim holds. Every count-side number cited for mahadevpura-2 matches `output/local_vs_gemini_6.json` exactly (n_gem=39, local n=40, count_ratio 1.0256, matched=12, splits=4, merges=5, gem_rallies_in_merges=11, mean_iou=0.4205), and the "near-perfect count masks poor boundary localization" interpretation is well-supported ? the right count is a coincidence of offsetting over-splits and over-merges, while only 12/~40 rallies cleanly match at a mediocre 0.42 IoU. The 40-owner-label ground truth (mean 5.66s) is corroborated by `output/mahadevpura_2_golden_features.json`. The F1@0.5 figures (0.150 / 0.296) are not in the cited JSON but are not required to substantiate the claim.